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NIM : E41230403 / A

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Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Acer> winget install Kubernetes.minikube
The `msstore` source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function prope
rly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes  [N] No: Y
Found Kubernetes - Minikube - A Local Kubernetes Development Environment [Kubernetes.minikube] Version 1.38.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/kubernetes/minikube/releases/download/v1.38.1/minikube-installer.exe
  48.5 MB / 48.5 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\Acer>
```

```
Windows PowerShell   Windows PowerShell

PS C:\Users\Acer> minikube start --driver=docker
😄  minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
✨  Using the docker driver based on user configuration

💣  Exiting due to PROVIDER_DOCKER_VERSION_EXIT_1: "docker version --format <no value>-<no value>:<no value>" exit statu
s 1: failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and
if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
📘  Documentation: https://minikube.sigs.k8s.io/docs/drivers/docker/

PS C:\Users\Acer> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\Acer> minikube start --driver=docker
😄  minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
✨  Using the docker driver based on user configuration
❗  Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
📌  Using Docker Desktop driver with root privileges
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
💾  Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 2.49 Mi
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 3.33 Mi
🔥  Creating docker container (CPUs=2, Memory=4000MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\Acer>
```

```
PS C:\Users\Acer> kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   2m31s   v1.35.1
PS C:\Users\Acer>
```

```
PS C:\Users\Acer> minikube start --driver=docker
😄  minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
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PS C:\Users\Acer> kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   2m31s   v1.35.1
PS C:\Users\Acer> ^C
PS C:\Users\Acer> cd ~
PS C:\Users\Acer> mkdir kubernetes


    Directory: C:\Users\Acer


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         4/29/2026   2:49 PM                kubernetes


PS C:\Users\Acer> cd kubernetes
PS C:\Users\Acer\kubernetes> code .
```

service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        imagePullPolicy: IfNotPresent
        ports:
        - containerPort: 80
```

```
PS C:\Users\Acer\kubernetes> kubectl apply -f deployment.yaml
deployment.apps/web-nginx configured
PS C:\Users\Acer\kubernetes> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-fc4c756f9-l68fq     1/1     Running   0          3s
web-nginx-fc4c756f9-m7glc     1/1     Running   0          5s
web-nginx-fc4c756f9-nc569     1/1     Running   0          2s
```

```
PS C:\Users\Acer\kubernetes> kubectl scale deployment web-nginx --replicas=5
deployment.apps/web-nginx scaled
PS C:\Users\Acer\kubernetes> kubectl scale deployment web-nginx --replicas=3
deployment.apps/web-nginx scaled
PS C:\Users\Acer\kubernetes>
```

```
PS C:\Users\Acer\kubernetes> kubectl get deployment -w
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx   3/3     3            3           9m27s
```

```
PS C:\Users\Acer\kubernetes> minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it might cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔥  Creating docker container (CPUs=2, Memory=2200MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
🏄  Successfully added m02 to minikube!
PS C:\Users\Acer\kubernetes> kubectl get nodes
NAME           STATUS     ROLES           AGE   VERSION
minikube       Ready      control-plane   20m   v1.35.1
minikube-m02   NotReady   <none>          57s   v1.35.1
PS C:\Users\Acer\kubernetes>
```

```
PS C:\Users\Acer\kubernetes> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running

PS C:\Users\Acer\kubernetes>
```

```
PS C:\Users\Acer\kubernetes> kubectl delete -f deployment.yaml
deployment.apps "web-nginx" deleted from default namespace
PS C:\Users\Acer\kubernetes> kubectl delete -f service.yaml
service "web-nginx-service" deleted from default namespace
PS C:\Users\Acer\kubernetes>
```